

# Class 5: Integrated Sensing and Communication

What it is, why it is hard, and how OFDM does it

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CIS 4930/5930 Future Edge Networks

[xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks](https://xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks)

# Two machines, one sky



Frankfurt airport surveillance radar

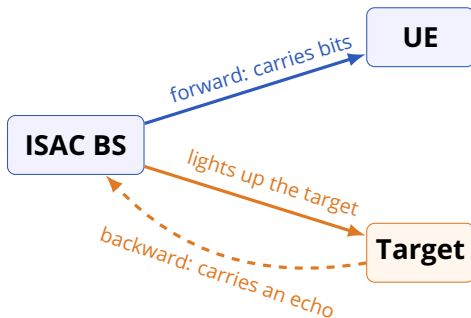


Ericsson active 5G antennas

**Q: One sends a wave to be understood. One sends a wave to come back. What do they share?**

**A:** Spectrum, power, antennas, and silicon — all of it scarce. Building two systems that never talk to each other pays for the same hardware twice.

# One waveform, two jobs



The same transmission is **information** on the way out  
and **measurement** on the way back.

ISAC does not add a radar to the network.  
It reads the network's own signal as radar.

# An echo is data nobody uploaded



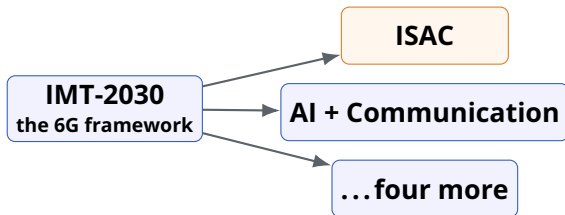
Waymo Jaguar I-Pace roof sensor package

- **Vehicles:** see what the camera cannot.
- **XR:** track a hand, no wearable.
- **IoT:** count people, no camera.
- **Space:** range the satellite you talk to.
- **Safety:** a drone, a fall, a flood.
- **Farming, weather, health:** same echo.

**Every one of these is the network noticing something on its own.**

XR = Extended Reality    IoT = Internet of Things

# Sensing is written into the 6G frame



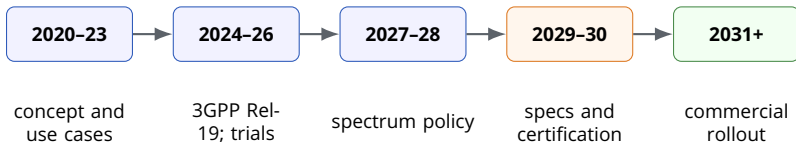
**Q: Why does a standards body have to say this out loud?**

**A:** Because a usage scenario is what gets spectrum, performance requirements, and test procedures written for it. Until ISAC was named, no radio was obliged to be measurable.

ITU-R Recommendation M.2160, IMT-2030 framework (2023)

# Being named starts a clock

Being written into M.2160 does not ship a product; it starts a schedule — spectrum, requirements, tests, then certification. Here is where that schedule stood when this material was prepared.

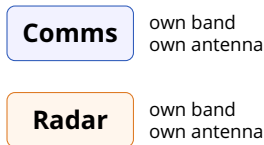


The 2020–2026 columns describe work that has happened. Everything from 2027 rightward is a forecast made in this talk, not a published schedule.

Standardisation is where the tradeoff stops being a curve and starts being a number somebody must certify.

# Two boxes became one box

## Separate systems



## DFRC



## What it buys

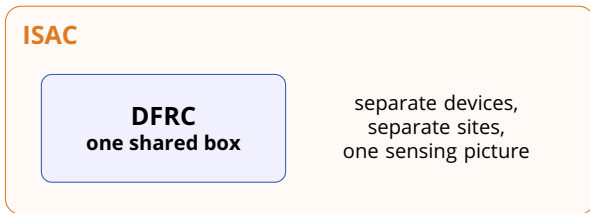
One spectrum licence, one amplifier, one aperture, one power budget.

DFRC = Dual-Function Radar Communication

## What it costs

A single waveform now has to satisfy two specifications at once.

# DFRC is one box. ISAC is the question.

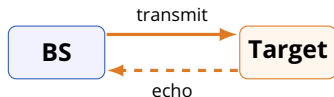


**DFRC** shares waveforms,  
antennas, and processing chains  
inside one radio.

**ISAC** also lets one node transmit  
and a different node listen.

Every DFRC system is ISAC.  
Not every ISAC system fits in one box.

# What an echo actually tells you



- **Delay** → range,  $R = c\tau/2$
- **Doppler** → closing speed
- **Angle** → direction

**Q: Which knob buys finer range, and which buys finer speed?**

**A: Bandwidth** buys range resolution — a wider signal has a sharper peak in time. **Observation time** buys velocity resolution — you must watch long enough for the phase to drift.

# Three questions for the rest of today

## 1 Why can't one signal be great at both jobs?

The two jobs want opposite things. That gives us a curve of compromises, not a fix.

## 2 Which signal do we actually use?

OFDM — the one Wi-Fi, 4G and 5G already transmit. No new radio.

## 3 How do we share it between bits and echoes?

The OFDM grid is a budget. Somebody has to decide who gets each cell.

Everything from here answers one of these three.  
If you lose the thread, ask which one we are on.

# They want opposite things (1/2)

|                 | Sensing wants                     | Communication wants                 |
|-----------------|-----------------------------------|-------------------------------------|
| <b>Spectrum</b> | Wide bandwidth, for resolution    | Efficient bandwidth, for rate       |
| <b>Signal</b>   | Deterministic waveforms           | Random, information-bearing symbols |
| <b>Power</b>    | High transmit power on the target | Power efficiency per bit            |
| <b>Time</b>     | Continuous illumination           | Low-latency bursts                  |
| <b>Space</b>    | Wide or scanning beams            | Narrow beams, one per user          |

Randomness is what carries information — and randomness ruins a clean echo.  
That single sentence is the whole conflict.

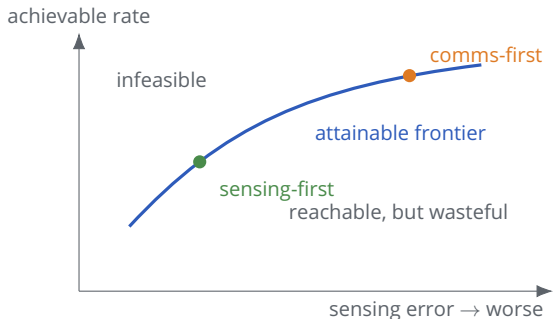
# They want opposite things (2/2)

| Sensing wants       |                                       | Communication wants                     |
|---------------------|---------------------------------------|-----------------------------------------|
| <b>Interference</b> | A clean echo, no clutter              | A clean constellation, no radar leakage |
| <b>Domain</b>       | Fine steps in distance and speed      | Fine slots in time and frequency        |
| <b>Hardware</b>     | Precise timing, high-power amplifiers | Spectral efficiency, many users         |
| <b>Latency</b>      | Low-latency tracking feedback         | Tolerates delay off the critical path   |
| <b>Regulation</b>   | Frequency allocation rules            | Emission and efficiency standards       |

**Q: Is there a setting where both are optimal?**

**A:** No. There is a **Pareto frontier**: past a point, every improvement in one is paid for out of the other. Design work is choosing where to sit on it, not escaping it.

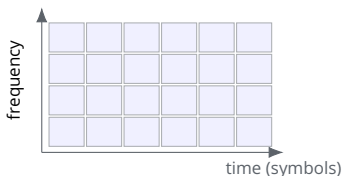
# The frontier, drawn once



Horizontal axis = sensing error: how far off the range and speed estimates are. **Lower is better**, so the good corner is up and to the left — and the curve is exactly what keeps you out of it: walking left buys accuracy with rate.

Every result in the rest of this lecture is a claim about one point on a curve like this.

# OFDM already carries a radar inside it



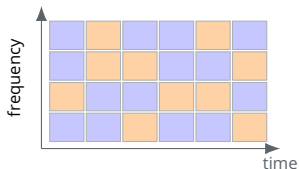
- Mature: Wi-Fi, 4G, 5G, 6G.
- The grid is already knobs.
- A cell serves bits or echoes.
- No new modulator needed.

**Q: If OFDM is already good enough, what is left to design?**

**A:** Which cells go to bits and which go to echoes. Every cell handed to sensing is a cell taken from data.

OFDM = Orthogonal Frequency Division Multiplexing

# The grid is the budget



■ communication RE   ■ sensing RE

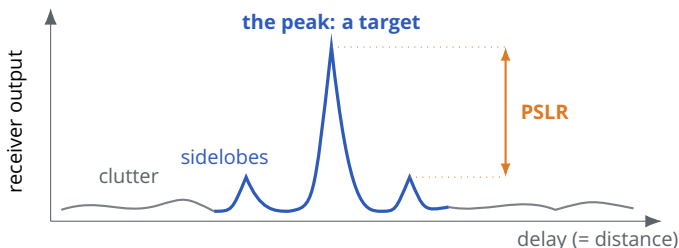
- One RE is one cell.
- Each carries symbol and power.
- Assign cell, then power.
- Two functions, one grid.

RE = Resource Element, one subcarrier during one OFDM symbol

PSLR = Peak-to-SideLobe Ratio; RoI = Region of Interest: the distances and speeds we actually care about

Cross-Domain Dual-Functional OFDM Waveform Design, IEEE JSAC 2024

# What a peak is, before we grade one



The receiver slides a copy of its own waveform across the echo. Where a target sits, everything lines up and adds — that spike is **the peak**. **Sidelobes** are ghosts of our own waveform; **clutter** is every echo we never asked for.

PSLR = how far the true peak stands above the tallest ghost.  
That number is the sensing grade on the next slide.

# Two scores, graded in two rooms

## Communication score

Achievable data rate, summed over every RE assigned to communication.

Measured in **time-frequency**: put bits where the channel is good.

## Sensing score

PSLR inside the region of interest — how far the true peak stands above the clutter.

Measured in **distance and speed**: make the peak stand tall where the targets actually are.

The same grid is graded twice,  
by two examiners in two different domains.

# Whoever picks first, wins

## Communication-centric

- 1 Give the good REs to bits.
- 2 Then tune leftover powers.
- 3 Sensing takes the remainder.

Rate stays optimal; PSLR is merely good inside the RoI.

## Sensing-centric

- 1 Lock the peak shape inside the RoI.
- 2 Then tune the rest of the grid.
- 3 Bits take the remainder.

PSLR is locally perfect; rate approaches, not reaches, optimal.

**Q: Both designs claim “almost no loss”. How can that be?**

**A:** Because the constraint is only enforced inside the **region of interest**. Narrow the room you must be perfect in, and the rest of the grid is free for the other function.

# Three things to take home

- 1 **One waveform, two jobs.** The signal a base station already sends comes back as an echo, and that echo is measurement nobody had to upload.
- 2 **The two jobs want opposite things.** Sensing wants a wide, steady, predictable signal; data wants an efficient, bursty, random one. So there is a curve of compromises, not a fix.
- 3 **On OFDM, picking a point on that curve is a scheduling decision.** Which cells of the grid carry bits, which carry echoes, and who picks first.

ISAC is not a new radio.

It is a new question asked of the radio we already have: who gets each cell?

# What is actually unfinished

## Settled enough to build

- The tradeoff is understood.
- OFDM-based ISAC works.
- The RE grid can be allocated.

## Still open

- A killer application.
- Security of a shared waveform.
- AI for waveform design.
- Space-air-ground reach.

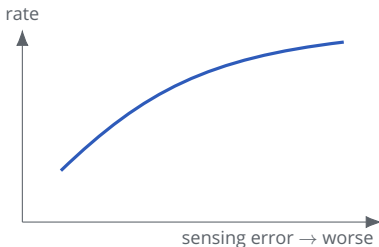
### **Q: Which of these is not an engineering problem?**

**A:** The first. A waveform can be optimised; a market cannot. ISAC has a technology looking for the application that pays for it.

SAGIN = Space-Air-Ground Integrated Network

# Before you go: pick a point on the curve

Two minutes, on paper, handed in on the way out. Same curve as before.



- 1 Mark where a **self-driving car** should sit. Mark where a **home Wi-Fi link** should sit.
- 2 Name one resource the two designs would allocate differently.
- 3 Answer in one sentence:  
**what does sensing cost the bits in your example?**

# Appendix: the acronym decoder ring

| Area     | Term     | Meaning / why it matters                                                                      |
|----------|----------|-----------------------------------------------------------------------------------------------|
| System   | ISAC     | Integrated Sensing and Communication                                                          |
| System   | DFRC     | Dual-Function Radar Communication; ISAC inside one radio                                      |
| System   | IMT-2030 | The ITU framework naming ISAC as a 6G usage scenario                                          |
| Waveform | OFDM     | Orthogonal Frequency Division Multiplexing                                                    |
| Metric   | PSLR     | Peak-to-SideLobe Ratio; true peak above the clutter                                           |
| Metric   | SINR     | Signal to Interference-plus-Noise Ratio                                                       |
| Resource | RE / RoI | Resource Element (one grid cell); Region of Interest (the distances and speeds we care about) |

# Appendix: the quantities behind every plot

**Delay**  $\tau$  range,  $R = c\tau/2$

**Doppler**  $\nu$  speed,  $v = \nu c/(2f_c)$

**Bandwidth** buys range resolution

**Dwell time** buys velocity resolution

**Rate**  $\sum \log_2(1 + \text{SINR})$  over REs

**Sensing error** has a hard floor set by bandwidth, power and time

**PSLR** peak over worst sidelobe, in the RoI

Rate is a sum you maximise.  
Sensing error has a floor you cannot go below.

Sensing accuracy is bounded from below.  
That is why the frontier bends.

- **Zhu Han (Univ. of Houston), “Integrated Sensing and Communication for 6G”**

The invited talk this lecture is built from. Slides and code: [wireless.egr.uh.edu](https://wireless.egr.uh.edu).

- **ITU-R M.2160: IMT-2030 framework**

Where ISAC becomes an official 6G usage scenario.

- **Cross-domain dual-functional OFDM waveform design**

IEEE JSAC 2024. Source of the RE-allocation results.